

Expanding, Factorising & Solving Equations

Quick reference sheet · GCSE Maths, Algebra · keep handy for revision

1. Expanding single brackets

$a(b + c) = ab + ac$ — multiply everything inside by what is outside. A negative multiplier flips every sign inside.

$$3(x + 4) = 3x + 12 \quad \cdot \quad -2(x + 6) = -2x - 12 \quad \cdot \quad -(3x - 5) = -3x + 5$$

2. Expanding double brackets (FOIL)

Multiply every term in the first bracket by every term in the second: **First, Outer, Inner, Last**.

Brackets	Expanded & simplified
$(x + 2)(x + 6)$	$x^2 + 8x + 12$
$(x - 2)(x + 7)$	$x^2 + 5x - 14$
$(x - 4)(x - 3)$	$x^2 - 7x + 12$

3. Factorising

Common factor: take out the highest common factor of every term. e.g. $6x + 9 = 3(2x + 3)$

Quadratic $x^2 + bx + c$: find two numbers that multiply to c and add to b .

$$x^2 + 7x + 12 \text{ — 3 and 4 multiply to 12, add to 7 } \rightarrow (x + 3)(x + 4)$$

$$x^2 + 2x - 15 \text{ — 5 and -3 multiply to -15, add to 2 } \rightarrow (x + 5)(x - 3)$$

4. Solving linear equations

Balance method: do the same operation to both sides until x is alone. Expand any brackets first; collect x terms onto one side if x appears twice.

$$3x + 7 = 22 \rightarrow 3x = 15 \rightarrow x = 5$$

$$2(x + 3) = 16 \rightarrow 2x + 6 = 16 \rightarrow 2x = 10 \rightarrow x = 5$$

$$5x - 3 = 2x + 9 \rightarrow 3x - 3 = 9 \rightarrow 3x = 12 \rightarrow x = 4$$

5. Check your work

After factorising, expand your answer back out — it should match the original expression. After solving an equation, substitute your value of x back in to check both sides are equal.